FitBuddy-AI Fitness Plan Generator using Gemini Models

Phase 8 – Project Demonstration

1. Introduction

The demonstration phase presents the completed FitBuddy project to the evaluator. The goal is not simply to show individual screens, but to demonstrate the complete workflow from user input to AI generation, database storage, feedback, and plan revision. A planned demonstration makes the presentation clear and reduces the chance of missing an important feature.

2. Demonstration Preparation

Before the demonstration, the project folder should be checked to ensure that the virtual environment, application files, AI modules, templates, static directory, and configuration are available. Required dependencies should be installed in the active virtual environment. The Gemini configuration required by the project should also be ready.

The server is started using Uvicorn with the application import path. The terminal should show that the server has started successfully and that application startup is complete.

3. Step-by-Step Demonstration

Step 1: Activate the virtual environment.

Step 2: Start the application with Uvicorn in reload mode.

Step 3: Open the local FitBuddy URL in a browser.

Step 4: Show the home page and explain the input fields.

Step 5: Enter a sample username, user ID, age, weight, goal, and intensity.

Step 6: Submit the form and explain that FastAPI receives the form data and the Gemini workout generator processes it.

Step 7: Show the generated workout plan and nutrition tip on the result page.

Step 8: Explain that the generated information is stored in the database.

Step 9: Enter feedback for the generated plan.

Step 10: Submit the feedback and show the revised plan produced by the update module.

Step 11: Open the administrator user view using the configured admin key and show stored user records.

Step 12: Demonstrate the deletion function only if required for evaluation.

Step 13: Open the health endpoint and show the service status.

Step 14: Open /docs and demonstrate the automatically generated FastAPI Swagger documentation.

4. What to Explain During Demonstration

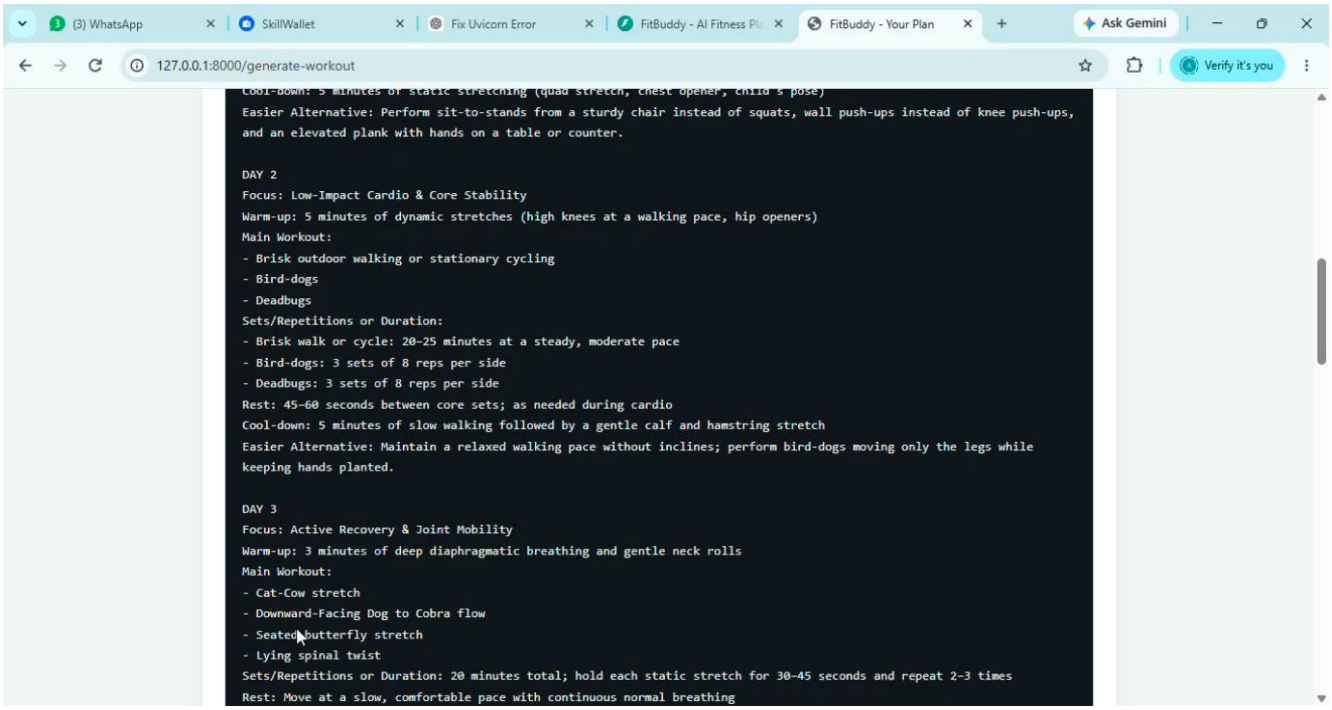
While showing the home page, explain that the interface collects the information needed for personalization. While showing the result page, explain that the workout is generated through Gemini rather than being a fixed response.

During the feedback demonstration, explain the role of the original plan, user feedback, goal, and age in the update process. When showing the database-related features, explain that the application uses SQLAlchemy with SQLite. When showing /docs, explain that FastAPI generates the API documentation automatically from the registered routes.

5. Expected Demonstration Output

A successful demonstration should show a working home page, successful form submission, AI-generated workout content, a nutrition tip, database persistence, feedback submission, an updated plan, administrator user viewing, and a successful health response. The exact AI-generated text may vary

because it is generated dynamically, but the application workflow should remain consistent.



6. Demonstration Troubleshooting

If the server does not start, first check that the virtual environment is activated and that Uvicorn is installed in that environment. If an import error appears, verify the module path and file location. If a template error occurs, verify that the templates directory and required HTML files exist and that Jinja2Templates points to the correct directory.

If form processing fails, check that python-multipart is installed. If AI generation fails, check the Gemini configuration, model availability, network connection, and quota. If the home page gives an internal server error, the terminal log should be checked immediately because it usually identifies the exact file and line where the request failed.

7. Conclusion

The demonstration phase proves that the FitBuddy project operates as an integrated system. By following a planned sequence, the team can show the

user interface, backend processing, Gemini AI integration, database storage, feedback-based updating, administration, and API documentation. This final phase connects all earlier phases and presents the completed project to the evaluator.

Phase Deliverables / Evidence

Item	Purpose	Status / Evidence
Demo sequence	Provides presentation order	Prepared
Feature walkthrough	Shows complete workflow	Prepared
Troubleshooting	Provides fallback checks	Prepared